Heart Disease Risk Prediction — MLOps Pipeline Report

Course: Machine Learning Operations (MLOps) AIMLCZG523 — Assignment 01 **Author:** Sahithi Siripuram **Repository:** https://github.com/SahithiSiripuram/heart-disease-mlops **Deployed API:** local Kubernetes (colima/k3s v1.35), 2 replicas behind a LoadBalancer service — reachable at `http://localhost:8080` via `kubectl port-forward svc/heart-disease-api 8080:80`

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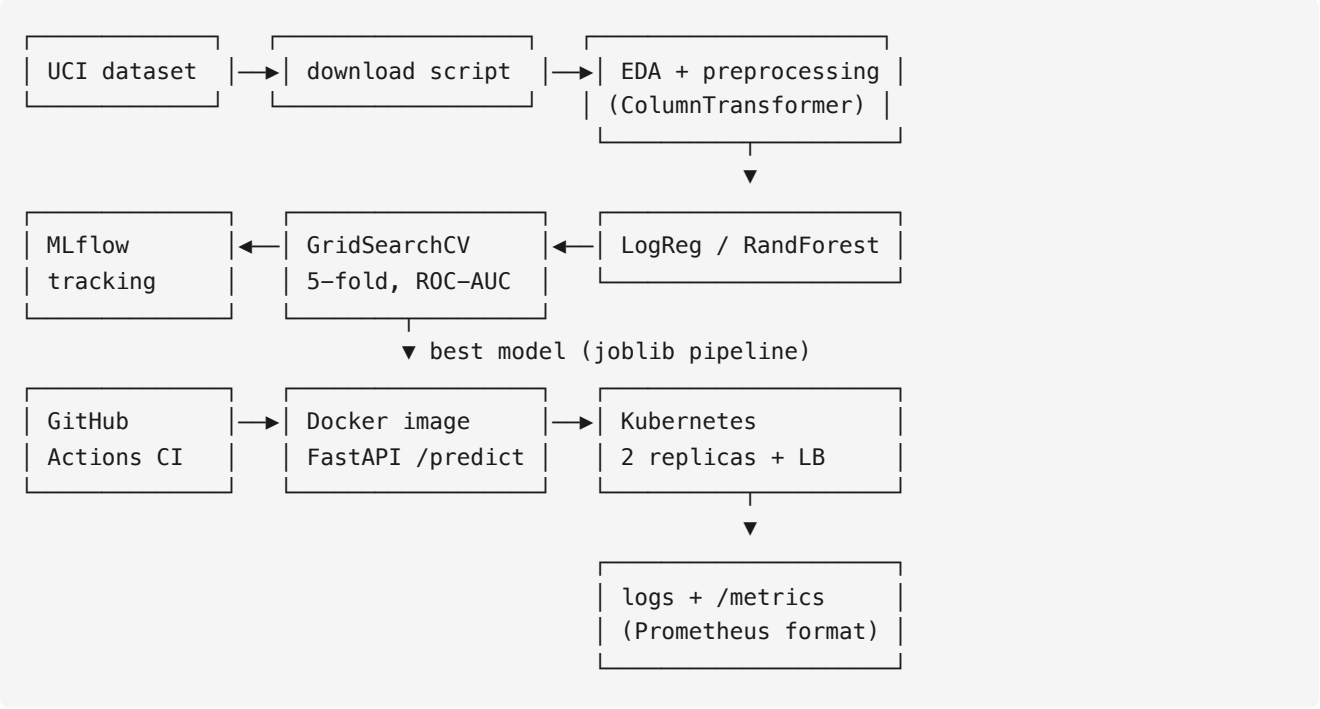
1. Project overview

This project builds a binary classifier that predicts the risk of heart disease from patient health data (UCI Heart Disease dataset) and delivers it as a production-ready, monitored API. The emphasis is on the operational pipeline: reproducible preprocessing, experiment tracking, automated testing and CI/CD, containerization, Kubernetes deployment, and monitoring.

Architecture



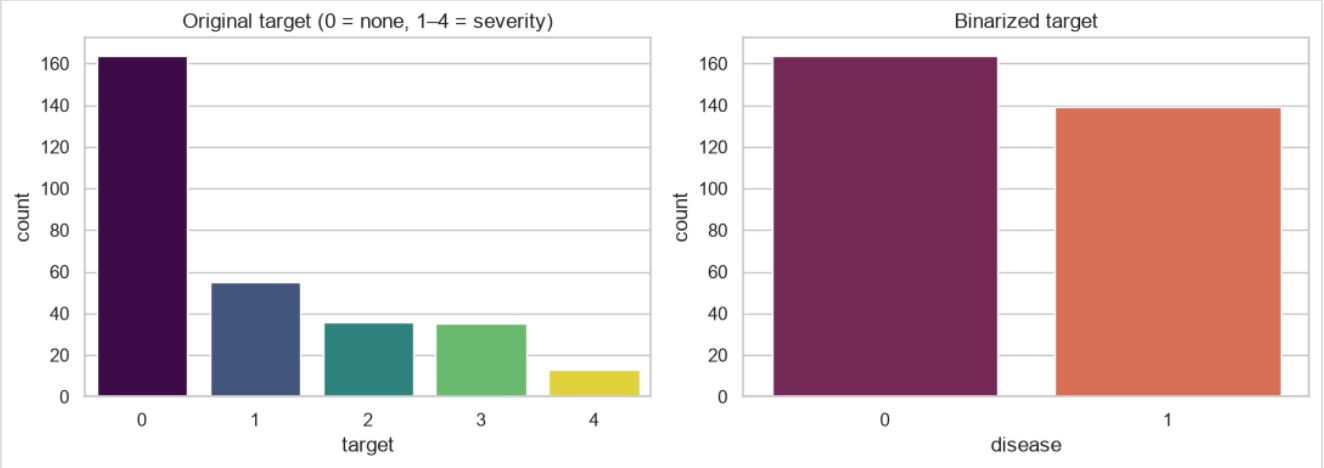
2. Data acquisition & EDA

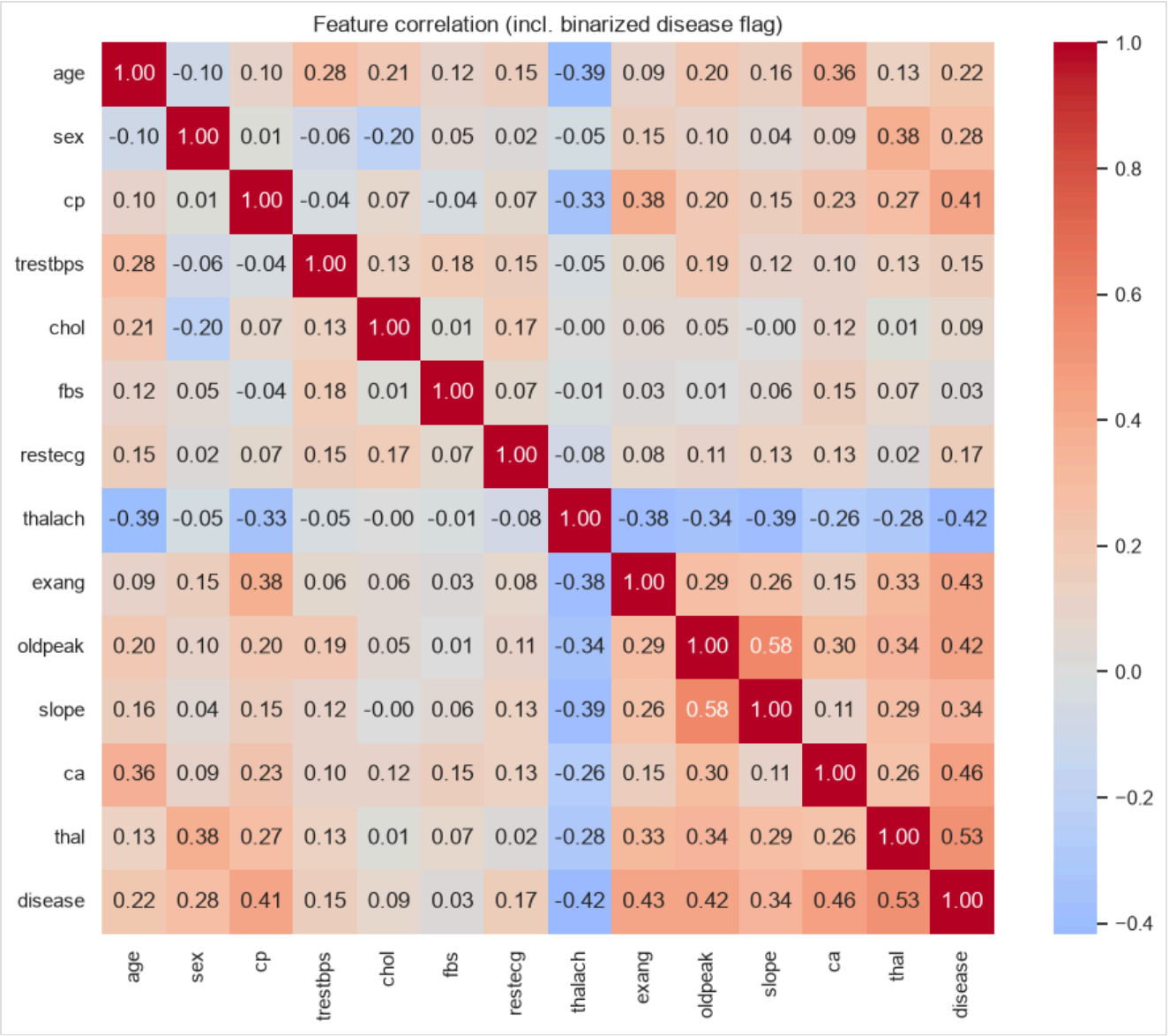
- **Source:** UCI ML Repository, Heart Disease dataset (Cleveland), fetched programmatically via `ucimlrepo` (`python -m src.data.download`); 303 rows, 13 features.
- **Target:** raw values 0–4 (0 = no disease, 1–4 = severity) binarized to presence/absence — classes are near-balanced (54% / 46%).

- **Missing values:** only `ca` (4 rows) and `thal` (2 rows) — imputed (median / most-frequent) instead of dropped.

Key EDA findings (full analysis in `notebooks/01_eda.ipynb`):

- Strongest signals: `thalach` (max heart rate, negatively associated), `oldpeak`/`slope` (exercise ST depression), `cp` (asymptomatic chest pain type 4 dominates the disease class), `ca`, `thal`, `exang`.
- Weak signals: `chol`, `fbs`, `trestbps`.
- Mixed numeric/categorical features motivate a ColumnTransformer with scaling (needed by logistic regression) and one-hot encoding.





3. Feature engineering & modelling

Preprocessing is a sklearn `ColumnTransformer` embedded in the model pipeline, so identical transforms run at training and inference:

- numeric (`age`, `trestbps`, `chol`, `thalach`, `oldpeak`, `ca`): median imputation + standard scaling
- categorical (`sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `thal`): most-frequent imputation + one-hot encoding (`handle_unknown="ignore"`)

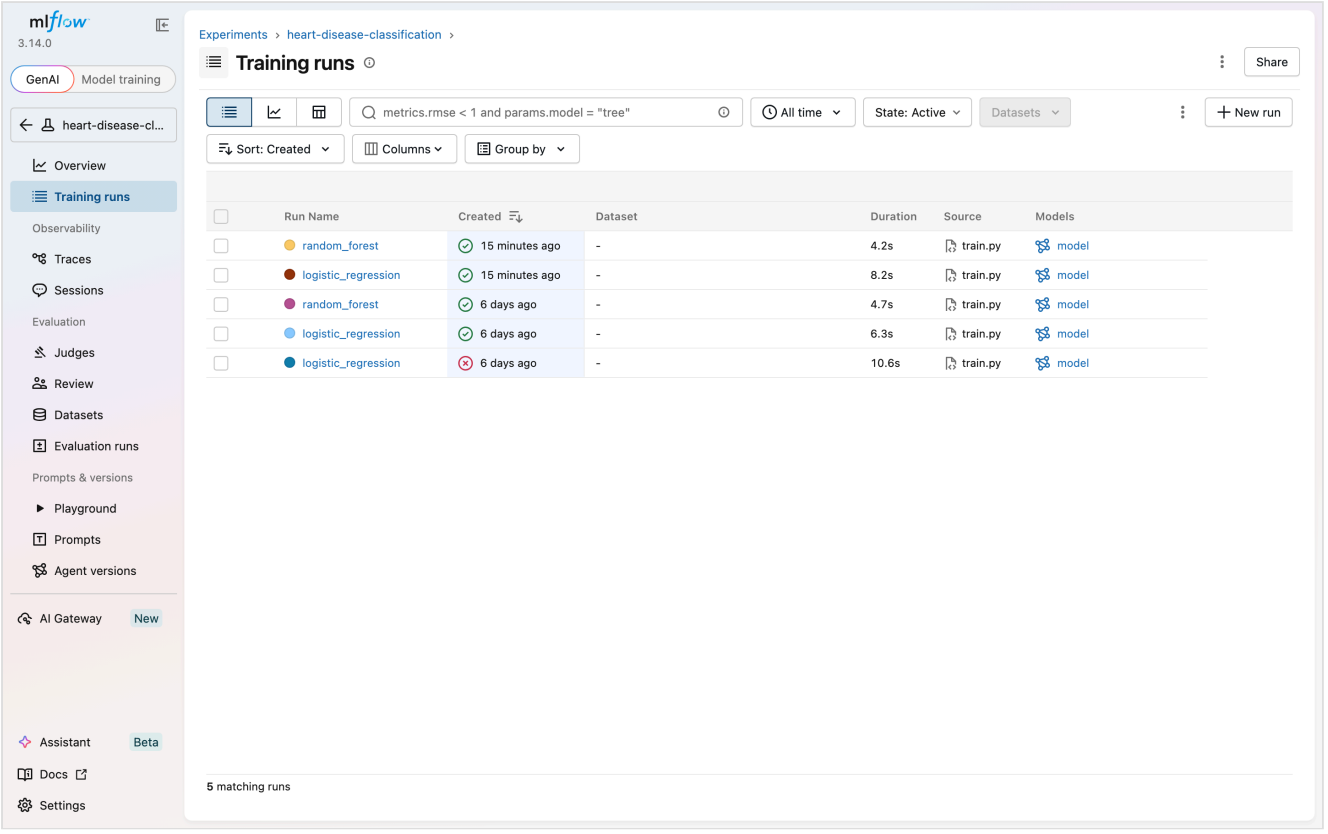
Two models were tuned with 5-fold GridSearchCV (scoring: ROC-AUC) over the full pipeline on an 80/20 stratified split:

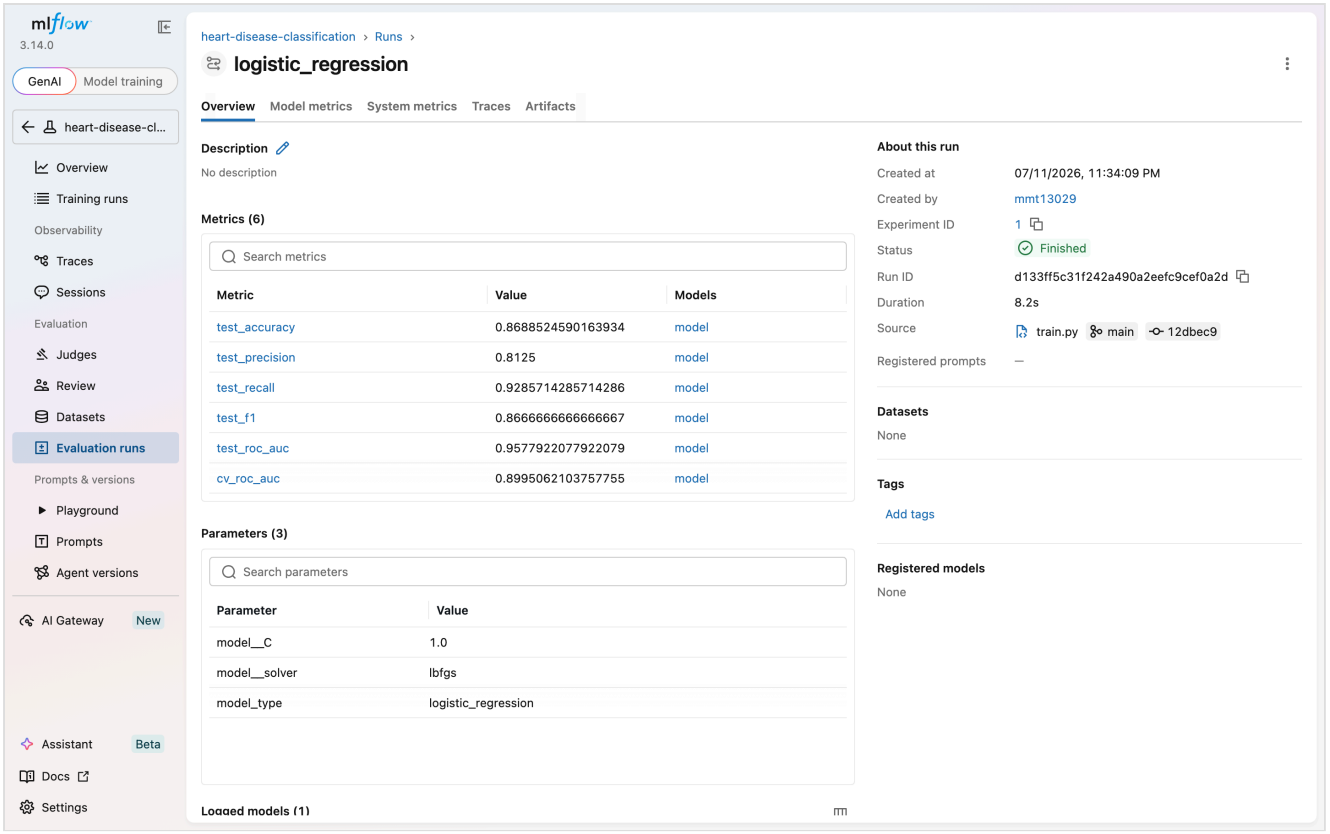
Model	Grid	Accuracy	Precision	Recall	F1	ROC-AUC (test)	ROC-AUC (CV)
Logistic Regression	C ∈ {0.01, 0.1, 1, 10}, solver ∈ {lbfgs, liblinear}	0.869	0.813	0.929	0.867	**0.958**	0.900
Random Forest	trees ∈ {100, 300}, depth ∈ {∞, 4, 8}, min-leaf ∈ {1, 3, 5}	0.885	0.839	0.929	0.881	0.943	0.900

Model selection: both models tie on cross-validated ROC-AUC (0.900); logistic regression was selected for its higher test ROC-AUC, simplicity, and interpretability — appropriate for a small (303-row) clinical dataset where a high-variance model gains little. Recall (0.929) matters most in this domain: missing an at-risk patient is costlier than a false alarm.

4. Experiment tracking (MLflow)

Every training run (`python -m src.models.train`) creates one MLflow run per model logging: best hyperparameters, all test metrics + CV ROC-AUC, the ROC curve and confusion matrix as figures, and the fitted pipeline as a model artifact.





5. Model packaging & reproducibility

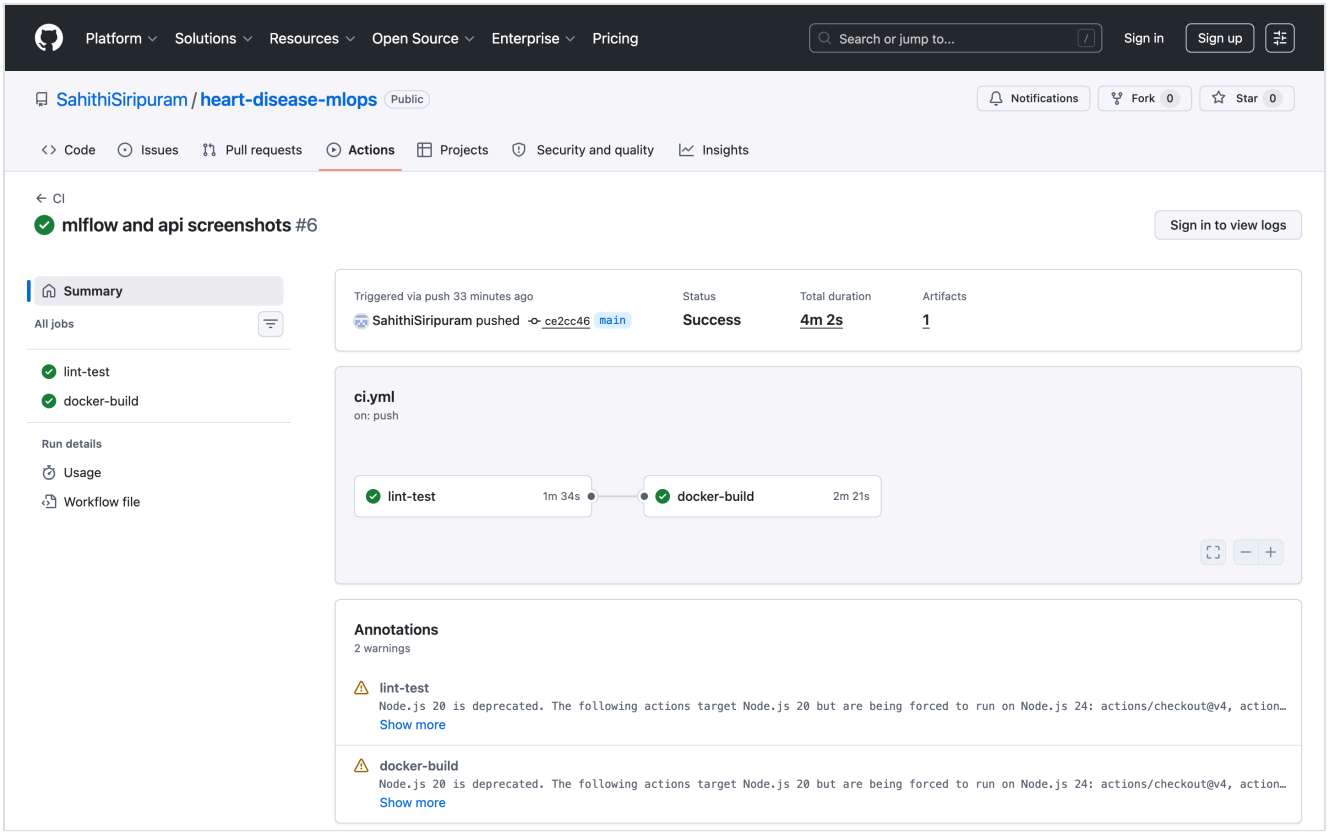
- Best pipeline exported to `models/model.joblib` (+ `model_metadata.json` with its metrics); models also stored per-run in MLflow.
- Clean-environment setup from `requirements.txt` alone; dataset fetched by script — no manual steps, verified in CI from scratch on every push.
- Fixed random seeds (splits, estimators) for reproducibility.

6. CI/CD (GitHub Actions)

`.github/workflows/ci.yml`, on every push/PR to `main`:

1. **lint-test job:** flake8 → pytest (8 tests: preprocessing behavior, API contract incl. validation errors) → dataset download → model training → upload of MLflow runs + model as build artifacts.
2. **docker-build job:** trains the model, builds the Docker image, boots the container, and smoke-tests `/health` and `/predict` with sample input.

The pipeline fails on any lint error, test failure, or build/smoke-test error.



7. Containerization & deployment

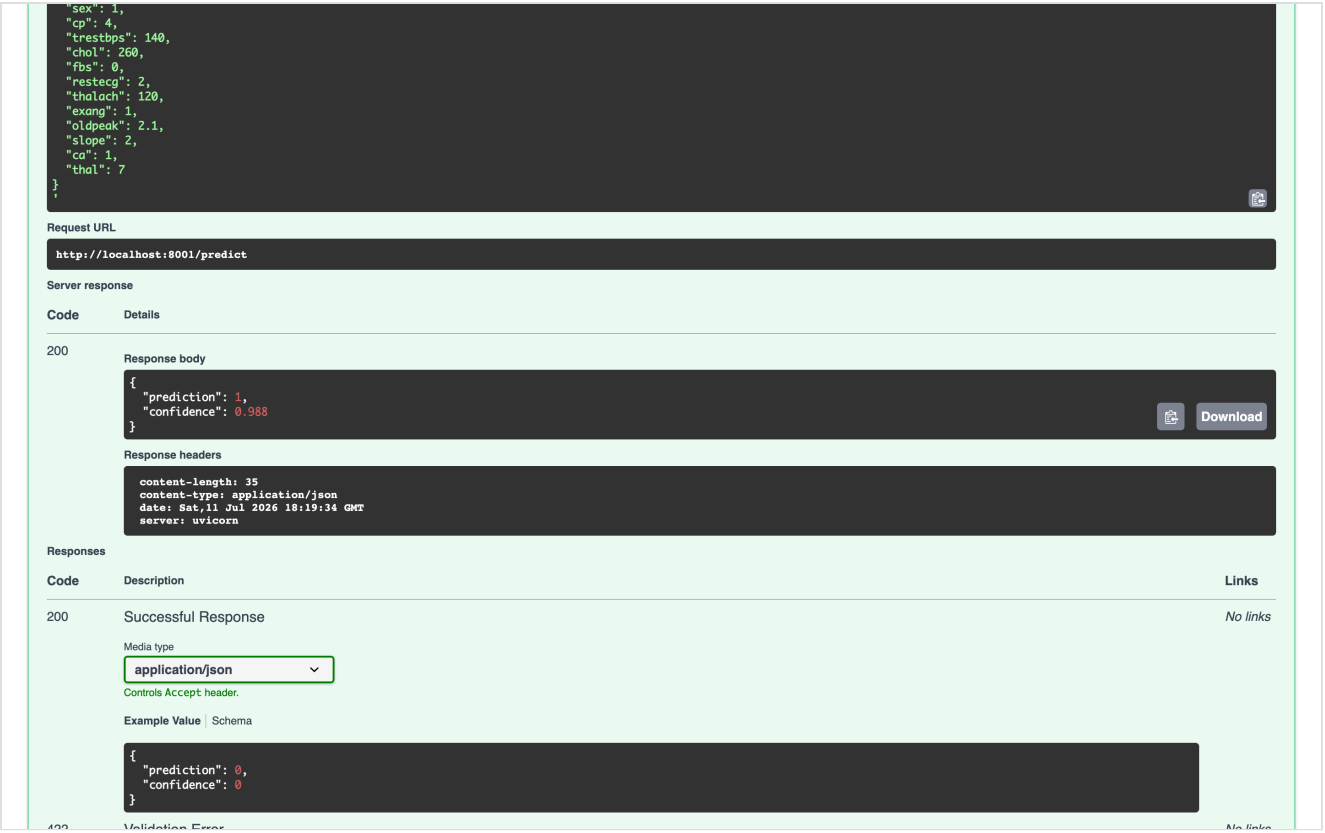
- **Docker:** `python:3.11-slim` base, dependencies from `requirements.txt`, serving via uvicorn on port 8000; `.dockerignore` keeps context minimal.
- **Kubernetes:** `deployment/deployment.yaml` (2 replicas, CPU/memory requests+limits, liveness/readiness probes on `/health`) and `deployment/service.yaml` (LoadBalancer → 8000).

Deployed on a local k3s cluster (colima). Both replicas pass their probes and the model answers through the cluster service; on colima the LoadBalancer's external IP is VM-internal, so the service is accessed from the host with `kubectl port-forward`:

```
$ kubectl get pods,svc -o wide
NAME                                    READY   STATUS    RESTARTS   AGE    IP          NODE     NOMINATED NODE   READINESS GATES
pod/heart-disease-api-5f94b8b947-gllkh   1/1     Running   0          2m1s   10.42.0.6   colima   <none>           <none>
pod/heart-disease-api-5f94b8b947-qcpv6   1/1     Running   0          2m1s   10.42.0.7   colima   <none>           <none>

NAME                       TYPE           CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE     SELECTOR
service/heart-disease-api   LoadBalancer   10.43.72.203   192.168.5.1   80:30973/TCP   2m1s    app=heart-disease-api
service/kubernetes          ClusterIP      10.43.0.1      <none>        443/TCP        4m14s   <none>

$ curl -X POST http://localhost:8080/predict -d @sample_input.json
{"prediction":1,"confidence":0.988}
```



8. Monitoring & logging

- Structured request logging middleware: method, path, status, latency per request.
- Prometheus metrics at `/metrics`: `api_requests_total` (by method/path/status), `api_request_latency_seconds` histogram, `predictions_total` (by class) — ready to scrape with Prometheus/Grafana.

```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 1273.0
python_gc_objects_collected_total{generation="1"} 40.0
python_gc_objects_collected_total{generation="2"} 26.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 374.0
python_gc_collections_total{generation="1"} 33.0
python_gc_collections_total{generation="2"} 3.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="12",patchlevel="12",version="3.12.12"} 1.0
# HELP api_requests_total Total API requests
# TYPE api_requests_total counter
api_requests_total{method="GET",path="/health",status="200"} 1.0
api_requests_total{method="POST",path="/predict",status="200"} 5.0
api_requests_total{method="GET",path="/docs",status="200"} 1.0
api_requests_total{method="GET",path="/openapi.json",status="200"} 1.0
# HELP api_requests_created Total API requests
# TYPE api_requests_created gauge
api_requests_created{method="GET",path="/health",status="200"} 1.783793728857016e+09
api_requests_created{method="POST",path="/predict",status="200"} 1.7837937300943239e+09
api_requests_created{method="GET",path="/docs",status="200"} 1.78379384374437e+09
api_requests_created{method="GET",path="/openapi.json",status="200"} 1.783793844346758e+09
# HELP api_request_latency_seconds Request latency
# TYPE api_request_latency_seconds histogram
api_request_latency_seconds_bucket{le="0.005",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.01",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.025",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.05",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.075",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.1",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.25",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.5",path="/health"} 1.0
api_request_latency_seconds_bucket{le="0.75",path="/health"} 1.0
api_request_latency_seconds_bucket{le="1.0",path="/health"} 1.0
api_request_latency_seconds_bucket{le="2.5",path="/health"} 1.0
api_request_latency_seconds_bucket{le="5.0",path="/health"} 1.0
api_request_latency_seconds_bucket{le="7.5",path="/health"} 1.0
api_request_latency_seconds_bucket{le="10.0",path="/health"} 1.0
api_request_latency_seconds_bucket{le="+Inf",path="/health"} 1.0
api_request_latency_seconds_count{path="/health"} 1.0
api_request_latency_seconds_sum{path="/health"} 0.0011520000116433948
api_request_latency_seconds_bucket{le="0.005",path="/predict"} 3.0
api_request_latency_seconds_bucket{le="0.01",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="0.025",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="0.05",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="0.075",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="0.1",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="0.25",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="0.5",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="0.75",path="/predict"} 4.0
api_request_latency_seconds_bucket{le="1.0",path="/predict"} 4.0
```

9. Setup instructions

See `README.md` for full commands. Summary:

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python -m src.data.download
python -m src.models.train
pytest && flake8 src tests
uvicorn src.api.main:app --reload
```

10. Repository link

https://github.com/SahithiSiripuram/heart-disease-mlops